

A GPS-Based Framework to Distinguish Road-Network Congestion from Bus Operation in Niterói, Brazil

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Public bus GPS data can serve as a citywide traffic probe, but slow bus movement often reflects transit operations rather than road-network congestion. Using the NetMob26 Data Challenge mobility dataset, this paper presents a GPS-based framework for distinguishing traffic-related slowdowns from bus-operational activity in Niterói, Brazil, using map-matched trajectories, rolling speed estimates, GTFS stop and terminal context, route information, and spatiotemporal clustering. Applied to a morning rush-hour case study in Centro, the framework retained 16,824 bus observations and identified 11,670 slow observations. Clustering results revealed persistent multi-route congestion events, including disruptions lasting over 30 minutes and involving up to 23 buses across 19 lines. The analysis also introduces a route-vulnerability measure showing that some routes are disproportionately exposed to road-network congestion, such as OC3, 57, 46, 40, and 32. Together, these results show that bus GPS data can support urban congestion detection while accounting for operational bias, and the framework can be generalized to other bairros/neighborhoods, dates, and time windows in cities where GPS, GTFS, and OSM data are available.

1. INTRODUCTION

Urban congestion has often been measured through general traffic sensors, probe vehicles, or aggregated road-speed platforms, but these data sources are not always available at the scale or resolution needed for citywide analysis. Travel-time data collection and floating-car methods have long treated moving vehicles as samples of network conditions rather than relying only on fixed roadside infrastructure [1, 2]. More recent probe-data approaches show that sparse, unevenly distributed vehicle trajectories can still support traffic speed estimation, though reconstructing network-wide traffic states from incomplete observations remains difficult [3]. Public buses offer a promising alternative because, unlike trains or trams, they are a transit service while still being subject to the urban road network.

Prior literature has shown that bus GPS, Automatic Vehicle Location (AVL), and General Transit Feed Specification (GTFS) Realtime data can be used as traffic probes for measuring arterial performance, identifying congestion hot spots, and estimating link-level speed variation [4–6]. Public buses are especially useful as probes because they operate throughout the day, repeatedly traverse important corridors, and often generate location observations of high-enough frequency to reveal short-duration slowdowns [5, 6]. Bus-probe studies have also been used to analyze travel-time variability and reliability, suggesting that repeated bus observations can reveal unstable road-network conditions when interpreted carefully [7, 8]. These studies motivate the use of public transit telemetry as a low-cost proxy for urban road-network performance.

However, buses are not passive observers of traffic. They are scheduled transit vehicles whose speeds are shaped not only by road-network conditions, but also by boarding and alighting, stop spacing, terminal layovers, route timing, signal delay, queueing behind other buses, and operational recovery time. Transit performance work using archived AVL/Automatic Passenger Counting (APC) data has emphasized that vehicle movement reflects both traffic conditions and transit-specific service behavior [9]. The *Transit Capacity and Quality of Service Manual* similarly treats dwell time, stop activity, route design, and passenger demand as central components of bus operating performance [10]. Treating all slow bus movement as congestion therefore risks turning the data into a “boy who cried wolf” story,

in which high congestion is repeatedly reported regardless of whether the slowdown reflects traffic, a passenger stop, or a terminal operation.

This paper therefore does not treat low bus speed as congestion by itself. This paper addresses that ambiguity by developing a GPS-based framework to determine if a slow observation is more plausibly a product of road-network congestion or bus operation. The framework combines bus GPS observations, OpenStreetMap (OSM) road-network geometry, GTFS stop and route context, rolling speed estimates, slow-speed scoring, and spatiotemporal clustering. GTFS is especially useful because it provides structured transit information such as stops, routes, trips, stop times, and route shapes [11, 12]. By comparing slow GPS observations with stop and terminal locations, the framework separates slow movement that is plausibly operational from slow movement that is more likely caused by road-network conditions.

This paper makes three main contributions. First, it proposes a framework for using bus GPS data as traffic probes while accounting for transit-operational bias. Second, it introduces a spatiotemporal clustering workflow that separates road-network clusters, stop/terminal clusters, and mixed or uncertain clusters. Third, it defines a route vulnerability ratio that identifies routes disproportionately represented in road-network congestion clusters relative to their overall slow-point activity.

2. RELATED WORK

Probe-vehicle methods infer road-network conditions from vehicles moving through the system. Floating-car data and other probe-vehicle techniques can estimate speed, travel time, and congestion formation, but these methods depend on where probe vehicles travel and how frequently observations are available [2, 3]. Stoll et al. used high-resolution bus GPS data to visualize and identify congestion hot spots on urban arterials, while Almeida et al. used bus tracking data to characterize urban traffic congestion patterns [4, 5]. Jiang et al. further showed that GTFS-Realtime bus data can support real-time urban traffic monitoring when bus observations are processed into road-segment traffic information [6].

At the same time, bus probe data have a distinct interpretive problem. A bus trajectory is both a traffic trace and a transit-

service trace. Travel-time variability in bus data can reflect congestion, but it can also reflect passenger demand, dwell time, route design, scheduled recovery, and operational control [7, 8].

3. DATA AND PREPROCESSING

This study uses the NetMob26 Data Challenge dataset for Niterói, Brazil, which includes high-resolution public bus mobility data, ticketing records, auxiliary transit information, and urban context data [13]. Telemetry data tracked bus GPS observations all hours of the day from March 11-31, 2026, with timestamps spaced ~ 15 s from each other. The analysis here focuses on the mobility and auxiliary transit components such as bus GPS observations, route labels, GTFS feeds, stop and terminal locations, and route context. Ticketing and ridership data are not used directly in this framework, but they motivate future passenger-weighted extensions of the vulnerability metric.

Census-sector, neighborhood, and municipal boundary layers for Niterói were obtained from the Instituto Brasileiro de Geografia e Estatística (IBGE), using the 2022 Census Sector Mesh and 2025 Municipal Digital Mesh products for the state of Rio de Janeiro [14, 15]. Niterói-specific bairro boundaries were derived by isolating the Niterói municipality from the Rio de Janeiro municipal boundary layer and using it to subset the IBGE 2022 intramunicipal bairro geometries. From there, we could focus our studies to any target bairro we desired. We decided to showcase the framework through a case study identifying observations from Centro between 07:30 and 08:30 AM on March 12, 2026. This time period was selected since it observed the highest amount of active trips in Niterói across the entire March 11-31 interval, as shown in Fig. 1. Centro was selected because it's a dense downtown environment with many overlapping bus routes, stops, terminals, and road-network bottlenecks. A simple low-speed map would mix all of these causes together, but the contribution of this paper is to make the distinction analytically explicit. If the method can separate operational activity from road-network congestion in Centro during morning rush-hour, it should be adaptable to less complex bairros where stops, terminals, and congestion are less spatially entangled. This, above all, is what makes the study area a useful stress test for the framework.

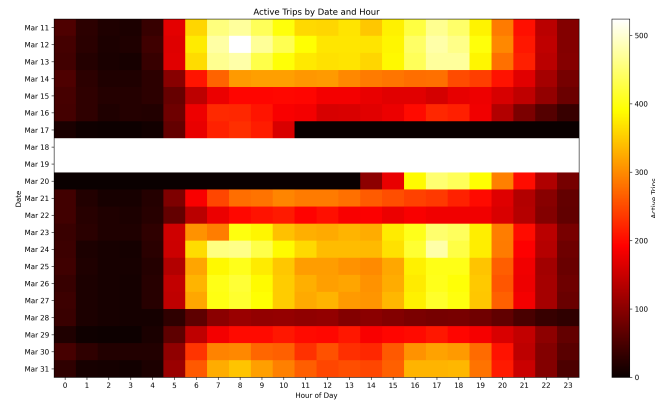


Fig. 1. A heatmap of active buses in the city of Niterói, Brazil, by hour of day.

Preprocessing consists of four operations. First, GPS observations are filtered spatially to the Centro study area and temporally to the morning rush-hour window. Second, road-network geometry is extracted from OpenStreetMap using OSMnx and projected into a metric coordinate system so that distances can be measured in meters [16, 17]. Third, GTFS stop, terminal,

and route information are loaded and reconciled with telemetry route labels. Fourth, observations are prepared for speed smoothing, map matching, stop/terminal distance calculation, and clustering. This preprocessing stage is necessary because the later classification equations depend on consistent spatial units, route labels, and stop/terminal geometry.

There are some irregularities in the NetMob26 Data Challenge dataset that can affect application of the framework and should be addressed. As can be easily observed from Fig. 1, telemetry data is completely nonexistent for March 18 and 19, along with an anomalous lack of active trips in parts of March 17, 20, and 28.

Missing vehicle probe data can crucially impact the accuracy of the results. As such, a survey of data completeness is obligatory for understanding dataset context, strengths, and weaknesses. This is also a factor in favor of the chosen case study setting, and usage of this framework, in Niterói or elsewhere, should be limited to time windows with sufficient telemetry coverage.

4. METHODOLOGY

A. Overview

The framework converts raw bus GPS observations into interpretable congestion events through six steps: map matching, rolling speed estimation, slow-score construction, GTFS-based labeling, clustering, and event linking.

Let each GPS observation be indexed by i , with timestamp t_i , projected latitude/longitude coordinates $\mathbf{x}_i = (x_i, y_i)$, observed speed v_i , smoothed speed $\tilde{v}_i$, route identifier r_i , and individual bus identifier b_i :

$$p_i = (t_i, \mathbf{x}_i, v_i, \tilde{v}_i, r_i, b_i) \quad (1)$$

The output of the method is a set of road-network congestion events, each containing a group of slow observations that are close in space and time and are not primarily explained by stop or terminal activity. The thresholds used throughout the framework are operational calibration parameters decided after experimentation. They were selected to balance GPS uncertainty, stop/terminal activity, and the need to identify multi-vehicle slowdowns, with stricter and more relaxed settings used as sensitivity checks.

B. Road-Network Context

Raw GPS points are spatially noisy, especially in dense urban environments. Map matching assigns GPS observations to plausible road-network segments, allowing movement to be analyzed in relation to actual streets rather than raw coordinate clouds. Hidden Markov map-matching methods are widely used because they account for GPS noise and the connectivity of the road network [18].

OpenStreetMap provides a global source of road-network geometry, and the OSMnx Python library allows researchers to download, construct, project, analyze, and visualize street networks from OpenStreetMap data [16, 17, 19]. This makes it possible to extract the road network for any bairro in Niterói and assign bus observations to nearby road segments. Because OpenStreetMap is volunteered geographic information, network completeness and geometric quality can vary by place, and this remains a relevant limitation for map matching and distance-based filtering [20]. Even so, the map-matched points in Niterói, Brazil are well documented and detailed, providing a great spatial foundation for the full project.

C. GTFS and Transit-Operational Context

The General Transit Feed Specification (GTFS) is an open standard for publishing transit system information, including stops, routes, trips, stop times, calendars, and route shapes [11, 12]. GTFS data is commonly used for transit visualization, accessibility analysis, routing, and service planning. For our purposes, stops and terminals are treated as locations where slow bus movement may reflect passenger service or scheduled operations rather than road-network congestion.

This distinction is supported by transit operations literature. Bus dwell time depends on passenger boarding and alighting, stop design, vehicle design, fare collection, and clearance time [10, 21]. More recent work on dwell-time estimation has also emphasized that time spent near stops can include deceleration, passenger service, acceleration, and re-entry into traffic [22]. A broader review of dwell-time research also shows that stop activity is a persistent source of bus delay and travel-time variability across transit systems [23].

D. Map Matching

Each GPS observation p_i is assigned to the nearest road-network edge using a nearest-edge map-matching function

$$\epsilon_i^* = \arg \min_{\epsilon \in E} d(\mathbf{x}_i, \epsilon) \quad (2)$$

where $d(\mathbf{x}_i, \epsilon)$ is the projected distance from GPS point $\mathbf{x}_i$ to edge ϵ , and E is the set of road-network edges taken from OpenStreetMap [18, 19]. As shown in Fig. 2, for visualization purposes, the city limits are outlined using GeoPandas and the IBGE 2025 Municipal Mesh data [14, 16, 24].

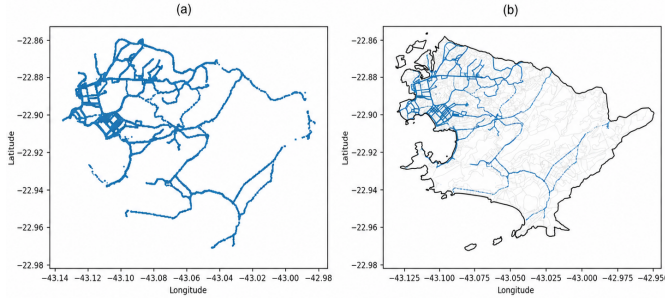


Fig. 2. (a) Raw GPS coordinates of every bus from March 11-31, 2026, taken from the NetMob26 mobility data. (b) The GPS coordinates from (a) mapped onto the city's OSM road-network edges, with the IBGE municipal boundary rendered using GeoPandas.

The snapped position of the observation is then denoted as

$$\mathbf{x}_i^* = P_{\epsilon_i^*}(\mathbf{x}_i) \quad (3)$$

where $P_{\epsilon_i^*}$ denotes projection onto the selected edge, as seen for all observation positions in Fig. 2.

E. Rolling Speed Estimation

Instantaneous GPS speeds can be noisy in the same way raw GPS observations are. Speeds can fluctuate because of sampling irregularity, GPS error, or short stop-and-go movements. To reduce this noise, speeds are smoothed within each bus trajectory. For each bus b , observations are sorted by timestamp and indexed in that order as $i_{b,1}, \dots, i_{b,n_b}$. If observation $i = i_{b,q}$, its smoothed

speed is computed from the current observation and up to three immediately preceding observations:

$$\tilde{v}_i = \frac{1}{|\mathcal{W}_i|} \sum_{j \in \mathcal{W}_i} v_j, \quad (4)$$

where

$$\mathcal{W}_i = \{i_{b,\ell} : \max(1, q-3) \leq \ell \leq q\}, \quad i = i_{b,q}. \quad (5)$$

A four-observation rolling window was selected experimentally. With the NetMob26 timestamps spaced approximately 15 s apart, a full window spans about 45 s from its earliest to latest observation while incorporating four consecutive samples. This suppresses frame-to-frame GPS-speed noise without averaging over long periods that could obscure short-duration stop-and-go structure. Near the beginning of a bus trajectory, fewer than four observations are available. Low-frequency AVL research has noted that bus location data often contains GPS or recording errors but that even 60-second vehicle data can remain statistically useful [25].

F. Slow-Score Construction

To determine whether a bus is moving unusually slowly, each smoothed speed is compared with a local free-flow reference speed. Let $v_{\min} = 1$ km/h be the stopped-vehicle threshold. Speeds below this threshold are basically stationary for vehicle movement, but they can account for slight GPS irregularity. Speeds less than $v_{\min}$ are excluded from the free-flow calculation so that stationary buses do not bias the reference downward. The free-flow reference follows the same logic as congestion-reporting practice in which free-flow speed is associated with high-percentile, relatively uncongested speeds, although here the percentile is computed from the retained bus observations in the local study window [26].

The free-flow speed is defined as the 85th percentile of smoothed moving speeds:

$$v_{\text{ff}} = Q_{0.85}(\{\tilde{v}_i : \tilde{v}_i > v_{\min}\}). \quad (6)$$

This gives a local operational free-flow speed for the study area and time window. In the Centro case study, this value was estimated as $v_{\text{ff}} = 19.70$ km/h. This relatively low value is expected in this case and reflects the dense downtown operating environment rather than unconstrained travel through the road network.

Each observation is assigned a normalized slow-speed score measuring its deviation from the free-flow reference:

$$s_i = \max\left(0, 1 - \frac{\tilde{v}_i}{v_{\text{ff}}}\right). \quad (7)$$

Values near 0 indicate movement close to free-flow speed, while values near 1 indicate stopped or near-stopped movement. A slow observation is then defined using a slow-speed score threshold σ_s :

$$D_i^{\text{slow}} = \mathbf{1}[s_i \geq \sigma_s]. \quad (8)$$

In the Centro case study, the threshold was experimentally determined to be $\sigma_s = 0.48$, meaning that buses had to be moving at 52% or less of the local free-flow speed. This follows heavier-congestion conventions with speeds falling below about 50-60% of free-flow [27]. As such, σ_s is not intended to mark any deviation from free-flow travel, but rather to retain observations representing substantial slow movement suitable for spatial clustering.

Thus, only observations satisfying the slow-score criterion, $D_i^{\text{slow}} = 1$, are passed to the GTFS classification and clustering stages. All subsequent methodology therefore assumes that observations have sufficiently high slow-speed scores.

G. GTFS-Based Labeling

To separate likely operational slowdowns from road-network slowdowns, each slow observation is compared with its distance from a GTFS stop or terminal location. Using the snapped position from Eq. (3), the nearest stop and terminal distances are

$$d_i^{\text{stop}} = \min_{s \in S} d(\mathbf{x}_i^*, s), \quad d_i^{\text{term}} = \min_{\tau \in T} d(\mathbf{x}_i^*, \tau). \quad (9)$$

where S is the set of GTFS stops and T is the set of terminal locations. For observations already identified as slow, the point-level label is assigned by the rule

$$L_i = \begin{cases} \text{road-network,} & d_i^{\text{stop}} > \delta_{\text{stop}} \wedge d_i^{\text{term}} > \delta_{\text{term}} \\ \text{stop/terminal,} & d_i^{\text{stop}} \leq \delta_{\text{stop}} \vee d_i^{\text{term}} \leq \delta_{\text{term}} \end{cases} \quad (10)$$

The threshold distances were set to be $\delta_{\text{stop}} = 25\text{m}$ and $\delta_{\text{term}} = 60\text{m}$ respectively, following conventional thresholds for traffic stopping [28]. Thus, a slow observation is treated as road-network-related only when it is a sufficient distance δ_{stop} and δ_{term} away from both stops and terminals. These buffers are intentionally conservative: they remove slow observations occurring immediately at known operational locations, but they do not assume that all slow movement near a stop is purely operational. Ambiguous cases are therefore left to the cluster-level classification step, where route diversity, bus diversity, and dwell composition are considered jointly.

If it lies near either a stop or a terminal, it is classified as operational. This reflects the fact that near-stop delay may include deceleration, boarding, alighting, clearance, acceleration, and re-entry into traffic, rather than road congestion alone [21–23]. The label L_i is not the final congestion event classification; it is the observation-level input used in the cluster-level classification below.

H. Clustering

After slow observations are labeled as either road-network or stop/terminal candidates, they are grouped using frame-wise spatial DBSCAN to determine road-network congestion over time. DBSCAN is appropriate because it identifies dense groups of nearby slow observations, allows isolated points to remain as noise, and doesn't require the number of clusters to be specified in advance [29].

Let the analysis window run from time t_0 to t_f , and let Δt_{frame} denote the timestamp spacing. The total number of processing frames is

$$N_{\text{frame}} = \left\lceil \frac{t_f - t_0}{\Delta t_{\text{frame}}} \right\rceil. \quad (11)$$

Each frame is indexed by $k \in \{0, \dots, N_{\text{frame}} - 1\}$. For frame k , we define X_k as the ordered set of snapped locations of slow observations whose timestamps fall inside that frame:

$$X_k = \{\mathbf{x}_i^* : t_0 + k\Delta t_{\text{frame}} \leq t_i < t_0 + (k+1)\Delta t_{\text{frame}}\}. \quad (12)$$

DBSCAN is then applied separately to each frame, with parameters ε_m as the spatial neighborhood radius, and m_{min} as the minimum number of slow observations required to form a cluster. Because DBSCAN results depend on these parameters, the robustness of the output should be assessed through various sensitivity checks [30]:

$$\lambda_k = \text{DBSCAN}_{\varepsilon_m, m_{\text{min}}}(X_k). \quad (13)$$

DBSCAN returns a label vector λ_k with one entry for each observation in X_k . Each entry is a spatial cluster assignment which we shall call c . Non-negative integers such as $c = 0, 1, 2, \dots$ identify

DBSCAN cluster labels, while the value $c = -1$ indicates that the observation is treated as noise.

These DBSCAN labels are not the same as the GTFS-based point labels L_i . The DBSCAN label only describes whether that observation belongs to a dense spatial group in the current frame. Therefore, DBSCAN first determines the spatial cluster membership from Eq. (13), and the GTFS-based labels from Eq. (10) are used afterwards to classify the composition of each cluster.

For each non-negative DBSCAN label c appearing in frame k , define cluster C_{kc} as the set of observations in that frame assigned to label c . The road-network share r_{kc} is then computed as the fraction of observations in C_{kc} whose point-level label from Eq. (10) is "road-network":

$$r_{kc} = \frac{1}{|C_{kc}|} \sum_{i \in C_{kc}} \mathbf{1}[L_i = \text{road-network}]. \quad (14)$$

This share is used to assign cluster-level labels in much the same vocabulary as Eq. (10):

$$K_{kc} = \begin{cases} \text{road-network,} & r_{kc} \geq \rho \\ \text{stop/terminal,} & r_{kc} \leq 1 - \rho \\ \text{mixed/uncertain,} & 1 - \rho < r_{kc} < \rho \end{cases} \quad (15)$$

Here, ρ is an experimentally determined classification threshold. This frame-wise cluster-level classification prevents individual GPS points from being over-interpreted. A single slow point may be noisy or ambiguous, but a dense group of nearby slow points within the same frame provides stronger evidence of a meaningful slowdown pattern.

I. Event Linking

The frame-wise DBSCAN step identifies local, instantaneous concentrations of slow bus movement, but congestion episodes regularly persist over a longer period of time. Since DBSCAN is applied separately to each frame, the same physical slowdown may appear as a sequence of nearby road-network clusters over time. To recover these longer patterns, we link frame-wise clusters into slowdown events based on spatial and temporal continuity. This linking is applied to all cluster labels K_{kc} .

For each cluster C_{kc} , define its centroid as

$$\bar{\mathbf{x}}_{kc} = \frac{1}{|C_{kc}|} \sum_{p_i \in C_{kc}} \mathbf{x}_i^*. \quad (16)$$

Two clusters C_{kc} and $C_{k+1,c'}$ are linked when they have the same cluster-level label, occur in successive frames, and have centroids within the event-linking distance:

$$A_{cc'}^{(k)} = \mathbf{1}[K_{kc} = K_{k+1,c'}] \mathbf{1}[d(\bar{\mathbf{x}}_{kc}, \bar{\mathbf{x}}_{k+1,c'}) \leq \delta_{\text{event}}]. \quad (17)$$

The maximum spatial separation allowed between linked clusters is written as δ_{event} . The first condition requires the cluster-level labels from Eq. (15) to match frame-by-frame, while the second condition requires the clusters to occur in approximately the same location in the road network.

The set of connected clusters in these framewise adjacencies defines slowdown events. Each event is indexed as e , and $\mathcal{E}_e$ contains the cluster identifiers (k, c) belonging to event e . Because clusters are linked only when their cluster-level labels match, the event label is inherited consistently from its member clusters, $K_e = K_{kc}$ for any $(k, c) \in \mathcal{E}_e$. The full set of observations associated with event e is

$$\mathcal{O}_e = \bigcup_{(k,c) \in \mathcal{E}_e} C_{kc} \quad (18)$$

Event duration is measured as the number of frames spanned by the linked event, so we define the starting and ending frames as

$$k_0 = \min_{(k,c) \in \mathcal{E}_e} k, \quad k_f = \max_{(k,c) \in \mathcal{E}_e} k \quad (19)$$

The duration of event e is then the number of frames it spans multiplied by the timestamp spacing Δt_{frame} :

$$\text{dur}(e) = (1 + k_f - k_0) \Delta t_{\text{frame}}. \quad (20)$$

We also summarize each event by counting the distinct bus identifiers b_i and route identifiers r_i represented in $\mathcal{O}_e$. These event-level quantities describe the persistence, severity, and network relevance of each slowdown event while preserving the distinction between isolated frame-level clusters and longer slowdowns that persist across time. Lastly, the average slow-score character of each event is summarized by the mean slow-speed score over all observations in $\mathcal{O}_e$:

$$\bar{s}_e = \frac{1}{|\mathcal{O}_e|} \sum_{p_i \in \mathcal{O}_e} s_i. \quad (21)$$

J. Methodology Discussion and Detection Logic

This methodology is based on the idea that different slowdowns should leave different signatures in the data. A stop/terminal dwell event should be spatially close to a bus stop or terminal and include very low or zero speed. By contrast, a road-network congestion candidate should appear away from stops and terminals, persist across a road segment, and affect multiple vehicles or routes sharing the same roadway.

The distinction is not perfect. A congested road segment can be adjacent to a stop, and a stop can itself contribute to downstream congestion when buses block traffic or queue near the curb. The point of the framework is therefore not to claim perfect causal attribution for every slow GPS point. Instead, it makes the classification assumptions explicit and pushes ambiguous cases into the mixed/uncertain category. This is especially important in dense downtown areas like Centro, where terminal activity, passenger service, and road congestion are spatially entangled.

Detection stringency is controlled through several classification parameters and thresholds, including the slow-score threshold, DBSCAN radius, minimum cluster size, cluster-classification threshold, and event-linking distance. Lower stringency increases recall, but it also increases the risk of labeling operational activity as congestion. Higher stringency improves precision, but it may miss moderate congestion or events near stops. This tradeoff is a recurring issue in probe-data and density-clustering applications, where the observed signal is incomplete, unevenly sampled, and sensitive to parameter choices [3, 30].

5. CLUSTERING ANALYSIS

A. Centro Case Study Results

The framework was applied to Centro, Niterói, from 07:30 to 08:30 AM on March 12, 2026 (See Fig. 3). After map matching and GTFS context assignment, 16,824 bus observations were retained for clustering. Using the slow-score threshold $\sigma_s = 0.48$, 11,670 observations were classified as slow. Of these, 3,380 were labeled as stop/terminal dwell observations, while 8,290 were labeled as road-network slowdowns.

These counts show why operational filtering is necessary. A large share of the GPS signal reflects slow bus movement, but slow movement is not homogeneous. Some slow observations occur near stops and terminals, where dwell time and passenger service are expected [10, 22].

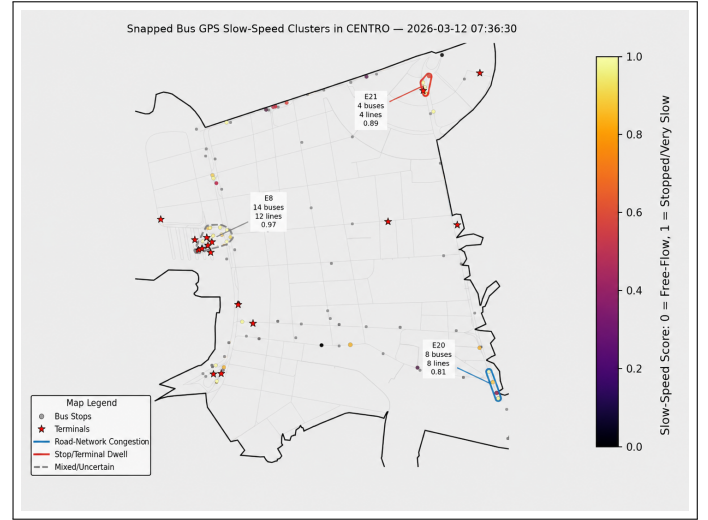


Fig. 3. Snapshot of example Centro transit data showing stop/dwell, road-network, and uncertain clusters.

DBSCAN clustering identified 634 road-network clusters, 179 stop/terminal clusters, and 94 mixed or uncertain clusters. The large number of road-network clusters suggests that slow bus movement in Centro is not limited to passenger-serving activity. At the same time, the substantial number of stop/terminal and mixed clusters confirms that operational behavior is an important part of the observed GPS signal.

The mixed category is especially important in a dense downtown. Stop activity, terminal proximity, and road congestion often overlap spatially. A bus may be delayed near a terminal that is also located on a congested road, or it may queue behind buses serving passengers. Forcing such cases into a binary road/operation classification would overstate certainty. The mixed label preserves that ambiguity.

B. Persistent Congestion Events

Table 1 shows that clustering events also identify persistent multi-route disruptions involving many buses at once. Event 31 lasted 33.50 minutes, involved up to 19 buses and 15 lines, and had a mean slow-speed score of 0.934. Event 73 lasted 25.25 minutes, involved up to 23 buses and 19 lines, and had a mean slow-speed score of 0.917.

Table 1. Clustering Event Results for Centro, March 12, 2026

Event Id	Duration (Min)	Max Buses	Max Lines	Slow Score
31	33.50	19	15	0.934
73	25.25	23	19	0.917
0	6.75	10	8	0.921
95	5.75	4	4	0.975
20	3.75	9	9	0.917

These events are stronger evidence of road-network congestion than isolated slow observations. A single slow bus could reflect stop service, driver behavior, or route-specific delay. Multiple buses from many routes moving slowly in the same area over an extended period are more likely to reflect a shared road-network disruption. This interpretation follows prior bus-probe studies in which repeated low-speed observations across vehicles are used to identify congestion hot spots and unstable travel-time conditions [4, 5, 7].

C. Interpreting Mixed Clusters

Mixed clusters are not simply failed classifications. In a dense downtown, they can reveal places where transit operations and road-network congestion interact. For example, a bus stop located on a heavily loaded corridor can produce operational dwell, but the same stopping activity may also contribute to queues that affect other buses and routes. Similarly, a terminal area may contain both scheduled layover activity and genuine congestion caused by many vehicles converging near the same access streets.

For this reason, the mixed/uncertain label is retained as a substantive result. It prevents the framework from overstating the precision of its classifications while still identifying locations that may deserve closer visual or operational inspection. In later applications, mixed clusters could be analyzed separately to identify areas where infrastructure design, stop placement, or terminal circulation contributes to network-level delay.

6. ROUTE VULNERABILITY ANALYSIS

A. Methodology

Route vulnerability measures whether a route is disproportionately represented in road-network congestion clusters relative to its overall slow-point activity. This extends prior AVL-based transit performance work from measuring service reliability to measuring route exposure to road-network congestion structures [9, 31].

Let R be the set of matched GTFS routes. For each route $r \in R$, define the number of slow observations from route r as

$$N_r^{\text{slow}} = \sum_i \mathbf{1}[r_i = r] D_i^{\text{slow}} \quad (22)$$

Let $\mathcal{C}^{\text{road}}$ denote the set of frame-wise clusters classified as road-network congestion clusters:

$$\mathcal{C}^{\text{road}} = \{(k, c) : K_{kc} = \text{road-network}\} \quad (23)$$

The number of route r observations appearing inside road-network congestion clusters is then

$$N_r^{\text{road}} = \sum_{(k,c) \in \mathcal{C}^{\text{road}}} \sum_{p_i \in C_{kc}} \mathbf{1}[r_i = r] \quad (24)$$

The corresponding totals across all matched routes are

$$N^{\text{slow}} = \sum_{r \in R} N_r^{\text{slow}}, \quad N^{\text{road}} = \sum_{r \in R} N_r^{\text{road}}. \quad (25)$$

Which defines the route vulnerability ratios as

$$V_r = \frac{N_r^{\text{road}} / N^{\text{road}}}{N_r^{\text{slow}} / N^{\text{slow}}}. \quad (26)$$

A route with $V_r > 1$ is over-represented in road-network congestion clusters relative to its overall slow-point activity. A route with $V_r < 1$ is under-represented. This distinction is important because a route with many buses may naturally produce many slow observations. The vulnerability ratio instead identifies routes whose slow observations are especially concentrated in road-network congestion clusters.

B. Route Performance

Several routes were strongly over-represented in road-network congestion clusters. The most notable example was OC3, as shown in Fig. 4. During the hour-long Centro case study, 78.6% of OC3 slow observations were located inside road-network clusters, and the route had a vulnerability ratio of 2.30. This

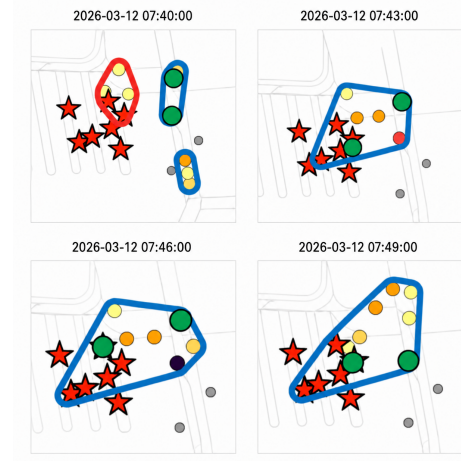


Fig. 4. Three-minute snapshots from Centro showing two OC3 buses, colored in green, with high route vulnerability because they remain within the same congestion event for an extended period. Map legend as in Fig. 3.

means that OC3 appeared in road-network congestion clusters more than twice as often as expected from its overall slow-point activity.

Other highly exposed routes included lines 57, 46, 40, and 32, all with vulnerability ratios above 1.5, as shown in Fig. 5. These routes were disproportionately associated with road-network congestion clusters. This suggests structural exposure to congested corridors or bottlenecks.

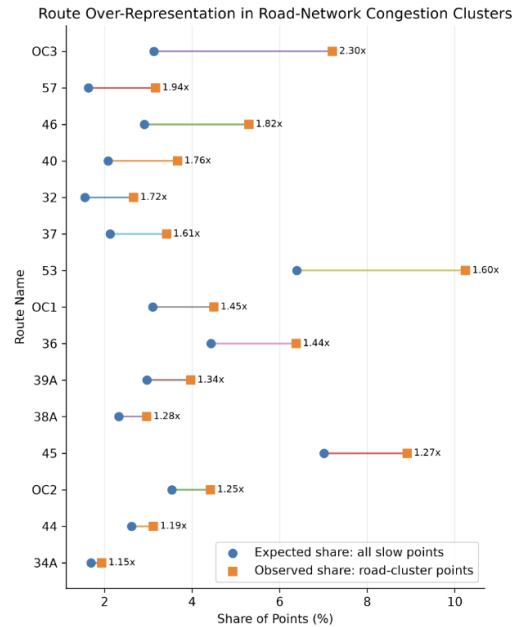


Fig. 5. A dumbbell chart depicting over-represented routes with vulnerability ratios above 1 in the Centro case study.

The vulnerability metric is useful because it normalizes for route activity. A frequent route may contribute many slow observations simply because it has many buses. The vulnerability ratio instead identifies routes whose slow observations are especially concentrated in road-network congestion clusters. This is conceptually similar to AVL-based transit reliability analysis, where vehicle location traces are used to identify route-level reliability problems, but the present metric focuses specifically

on over-representation inside road-network congestion clusters [9, 31]. In future work, this metric could be combined with ticketing or ridership data from the NetMob26 dataset to estimate passenger-weighted congestion exposure [13].

7. SENSITIVITY, VALIDATION, AND LIMITATIONS

The framework depends on several parameters: maximum snapping distance, rolling-window length, stopped-vehicle threshold, free-flow percentile, slow-score threshold, stop and terminal distance thresholds, DBSCAN spatial radius ϵ_m , minimum cluster size $m_{\min}$, cluster-classification threshold ρ , frame spacing Δt_{frame} , and event-linking distance δ_{event} . These parameters affect how many observations are retained, how many are classified as slow, how frame-wise clusters are formed, and how those clusters are linked into events.

The slow-score threshold σ_s is especially important. A lower threshold includes milder slowdowns, increasing recall but potentially adding noise. A higher threshold focuses on severe slowdowns but may miss moderate congestion. The free-flow percentile also matters because it determines the denominator of the slow-speed score; choosing a value too low would make slowdowns look less severe, while choosing a value too high could overstate congestion in a low-speed downtown network. The FHWA Urban Congestion Report's use of an 85th-percentile free-flow reference motivates the percentile approach used here, but the local bus-based reference should still be interpreted as an operational benchmark rather than a universal free-flow speed [26].

DBSCAN and event-linking parameters are also important because ϵ_m , $m_{\min}$, and δ_{event} control different stages of the detection process. If ϵ_m is too small or $m_{\min}$ too large, a spatial slowdown may fragment or be discarded; if ϵ_m is too large or $m_{\min}$ too small, distinct slowdowns may merge into the same frame-wise cluster. Separately, an overly small δ_{event} can split one persistent slowdown across successive frames, while an overly large value can link neighboring but distinct slowdowns. Prior DBSCAN work emphasizes both the strength of density-based clustering and the need to interpret its parameters in relation to the application domain [29, 30].

Validation is difficult because ground-truth congestion labels are rarely available at the scale of a full urban bus network. However, we used several indirect checks for verification. First, we visually compared cluster locations with animated GPS movement. Second, we used multi-route involvement as supporting evidence, since clusters involving many lines are more likely to reflect shared road-network conditions. Third, known terminal areas should produce stop/terminal clusters rather than false road-network clusters. Fourth, preliminary sensitivity checks using stricter and more relaxed parameter settings indicated that the largest events and highest-vulnerability routes were comparatively stable. These checks support the plausibility of the classifications but do not replace validation against independent ground-truth traffic data.

The main limitation is that buses are biased probes. Even after filtering out stop and terminal activity, buses do not experience the road network exactly like private vehicles. Their movement is constrained by routes, stop spacing, lane position, vehicle size, and passenger service [8, 10]. Additionally, nearest-edge snapping can misassign observations in dense networks, especially near intersections, terminals, or parallel streets [18]. Finally, OSM road-network quality may vary because OpenStreetMap is volunteered geographic information, so missing or simplified road geometry remains a possible source of error [19, 20]. The Centro case study covers one bairro and one hour, so the empir-

ical results should not be interpreted as a complete congestion profile for all of Niterói.

8. GENERALIZATION BEYOND CENTRO

Although the case study focuses on Centro, the framework is general. It can be applied to any bairro, date, or time window in any city where GPS, GTFS, and road-network data are available. The same workflow can be repeated: spatially filter observations, snap them to the OSM network, smooth speeds, estimate local free-flow speed, compute slow-speed scores, add GTFS context, classify slow observations, cluster them, link events, and compute route vulnerability.

Applying the method across multiple bairros would allow comparison of congestion structure across the city. Dense downtown areas may show many mixed clusters because stops, terminals, and traffic bottlenecks overlap. Peripheral bairros may show fewer terminal effects and more corridor-specific slowdowns. Applying the method across multiple dates would allow recurring congestion patterns to be separated from one-time disruptions. A road-network event that appears repeatedly during the morning peak would suggest a structural bottleneck, while a single-day event may reflect weather, a crash, construction, terminal disruption, or an unusual service pattern.

The route vulnerability metric can also be generalized. Comparing vulnerability across days could reveal whether certain routes are consistently exposed to road-network congestion or only affected during specific incidents. Combining route vulnerability with ridership data would make it possible to estimate not only which routes are exposed, but how many passengers are affected. Because NetMob26 includes multiple data sources beyond raw telemetry, the same framework could eventually support passenger-weighted exposure, bairro-level comparison, and citywide reliability diagnostics [13].

9. CONCLUSION

This paper developed a GPS-based framework for distinguishing road-network congestion from transit-operational slowdowns in public bus data. The central challenge is that buses are both traffic probes and transit vehicles. They observe congestion, but they also stop, dwell, queue, and recover schedule time for operational reasons.

By combining map-matched GPS trajectories, rolling speed estimates, GTFS stop and terminal context, spatiotemporal clustering, and route vulnerability analysis, the framework separates slow bus movement into more interpretable categories. The Centro case study showed that slow bus observations during the morning rush hour include road-network slowdowns, stop/terminal dwell activity, and mixed cases. Persistent road-network events involved many buses and lines, while route vulnerability analysis showed that some routes, including OC3, were disproportionately exposed to congestion.

The results demonstrate that bus GPS data can support urban congestion monitoring, but only when the operational nature of bus movement is explicitly accounted for. Future work should apply the framework across additional bairros, dates, and time windows; test parameter sensitivity more systematically; incorporate external validation data; and combine congestion exposure with passenger demand.

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Data availability. Data underlying the results presented in this paper is provided by the NetMob26 Data Challenge, for which access can be requested per the Data Challenge organizers, Ref. [13].

REFERENCES

1. S. M. Turner, W. L. Eisele, R. J. Benz, and D. J. Holdener, “Travel time data collection handbook,” Tech. Rep. FHWA-PL-98-035, Federal Highway Administration, U.S. Department of Transportation, Washington, DC (1998).
2. A. Kesting and M. Treiber, “Online traffic state estimation based on floating car data,” (2010).
3. F. Rempe, P. Franeck, and K. Bogenberger, “Estimating traffic speeds using probe data: A deep neural network approach,” (2021).
4. N. B. Stoll, T. B. Glick, and M. A. Figliozzi, “Using high-resolution bus GPS data to visualize and identify congestion hot spots in urban arterials,” *Transp. Res. Rec. J. Transp. Res. Board* **2539**, 20–29 (2016).
5. A. Almeida, S. Brás, S. Sargento, and I. Oliveira, “Exploring bus tracking data to characterize urban traffic congestion,” *J. Urban Mobil.* **4**, 100065 (2023).
6. S. Jiang, Y. Sun, W. Wong, *et al.*, “Real-time urban traffic monitoring using transit buses as probes,” *Transp. Res. Rec. J. Transp. Res. Board* **2679**, 219–236 (2025). First published online July 25, 2024.
7. N. Uno, F. Kurauchi, H. Tamura, and Y. Iida, “Using bus probe data for analysis of travel time variability,” *J. Intell. Transp. Syst.* **13**, 2–15 (2009).
8. E. Mazloumi, G. Currie, and G. Rose, “Using GPS data to gain insight into public transport travel time variability,” *J. Transp. Eng.* **136**, 623–631 (2010).
9. P. G. Furth, B. Hemily, T. H. J. Muller, and J. G. Strathman, “Using archived AVL-APC data to improve transit performance and management,” TCRP Report 113, Transportation Research Board, Washington, DC (2006).
10. Transportation Research Board, *Transit Capacity and Quality of Service Manual*, TCRP Report 165 (National Academies Press, Washington, DC, 2013), 3rd ed.
11. MobilityData, “General transit feed specification,” (2026). Accessed: 2026-07-06.
12. MobilityData, “GTFS schedule reference,” (2026). Accessed: 2026-07-06.
13. F. Domingos, H. T. Marques-Neto, B. Pereira, *et al.*, “The NetMob26 dataset: A high-resolution multi-source view of public bus mobility in Niterói,” (2026).
14. Instituto Brasileiro de Geografia e Estatística, “Malha Municipal Digital e Áreas Territoriais 2025,” (2026). Municipal boundary shapefile; Rio de Janeiro state file used: RJ_Municipios_2025.shp. Accessed 9 July 2026.
15. Instituto Brasileiro de Geografia e Estatística, “Malha de Setores Censitários 2022,” (2024). Geospatial vector files for census sectors and intramunicipal divisions; Rio de Janeiro state files used: RJ_setores_CD2022.gpkg and RJ_bairros_CD2022.gpkg. Accessed 9 July 2026.
16. G. Boeing, “OSMnx: A python package to work with graph-theoretic OpenStreetMap street networks,” *J. Open Source Softw.* **2**, 215 (2017).
17. G. Boeing, “OSMnx: New methods for acquiring, constructing, analyzing, and visualizing complex street networks,” *Comput. Environ. Urban Syst.* **65**, 126–139 (2017).
18. P. Newson and J. Krumm, “Hidden markov map matching through noise and sparseness,” in *Proceedings of the 17th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems*, (Association for Computing Machinery, New York, NY, USA, 2009), GIS '09, pp. 336–343.
19. M. Haklay and P. Weber, “OpenStreetMap: User-generated street maps,” *IEEE Pervasive Comput.* **7**, 12–18 (2008).
20. C. Barron, P. Neis, and A. Zipf, “A comprehensive framework for intrinsic OpenStreetMap quality analysis,” *Trans. GIS* **18**, 877–895 (2014).
21. K. J. Dueker, T. J. Kimpel, J. G. Strathman, and S. Callas, “Determinants of bus dwell time,” *J. Public Transp.* **7**, 21–40 (2004).
22. T. Alhadidi and H. A. Rakha, “Estimating transit vehicle dwell times at bus stops,” (2021).
23. S. Rashidi, S. Ataeian, and P. Ranjitkar, “Estimating bus dwell time: A review of the literature,” *Transp. Rev.* **43**, 32–61 (2023).
24. GeoPandas developers, “GeoPandas: Python tools for geographic data,” (2025).
25. Y. Yang, D. Gerstle, P. Widhalm, *et al.*, “Potential of low-frequency automated vehicle location data for monitoring and control of bus performance,” *Transp. Res. Rec. J. Transp. Res. Board* **2351**, 54–64 (2013).
26. Federal Highway Administration, “The urban congestion report: Documentation and definitions,” (2026). Accessed: 2026-07-06.
27. H. van der Loop, M. Kouwenhoven, and P. van Bekkum, “Validation of floating car data for transport policy analysis and transport models,” in *European Transport Conference 2017*, (Association for European Transport, 2017).
28. L. Killars, B. Schouten, and O. Mussmann, “Stop and go detection in gps-position data,” Discussion paper, Statistics Netherlands, The Hague, Netherlands (2019).
29. M. Ester, H.-P. Kriegel, J. Sander, and X. Xu, “A density-based algorithm for discovering clusters in large spatial databases with noise,” in *Proceedings of the Second International Conference on Knowledge Discovery and Data Mining*, (AAAI Press, 1996), KDD '96, pp. 226–231.
30. E. Schubert, J. Sander, M. Ester, *et al.*, “DBSCAN revisited, revisited: Why and how you should still use DBSCAN,” *ACM Trans. on Database Syst.* **42**, 1–21 (2017).
31. A. M. El-Geneidy, J. Horning, and K. J. Krizek, “Analyzing transit service reliability using detailed data from automatic vehicular locator systems,” *J. Adv. Transp.* **45**, 66–79 (2011).